

Interview



干皓丞

Interview

Before 2021 – Full-Stack Development and Data Analysis

Front-end and back-end development, data analysis, and Linux server maintenance.

Technologies: Angular, jQuery, JavaScript, PHP, R, CSS, HTML, and Linux Server Administration.

2021 年 之前 – 前後端開發跟數據分析

前後端開發、數據分析：Angular、Jquery、JS、PHP、R、CSS、HTML、Linux Server 維護

Since 2021 – Artificial Intelligence and Semiconductor Applications

Generative AI, medical image segmentation, industrial image segmentation, semiconductor defect inspection, automated visual inspection, and AI applications for manufacturing and process optimization.

2021 年 之後-人工智能

生成式人工智能、醫療影像分割、工業影像分割、YOLO、UNet、MambaUNet、RF-DETR、Pytorch、Python、PyQT、C#

Models and Technologies: YOLO, U-Net, Mamba U-Net, RF-DETR, PyTorch, Python, PyQt, and C#.

My current focus is on applying artificial intelligence and computer vision to semiconductor manufacturing, equipment automation, quality inspection, and production process improvement.



Work and Research Experience

My academic research focuses on medical image segmentation.

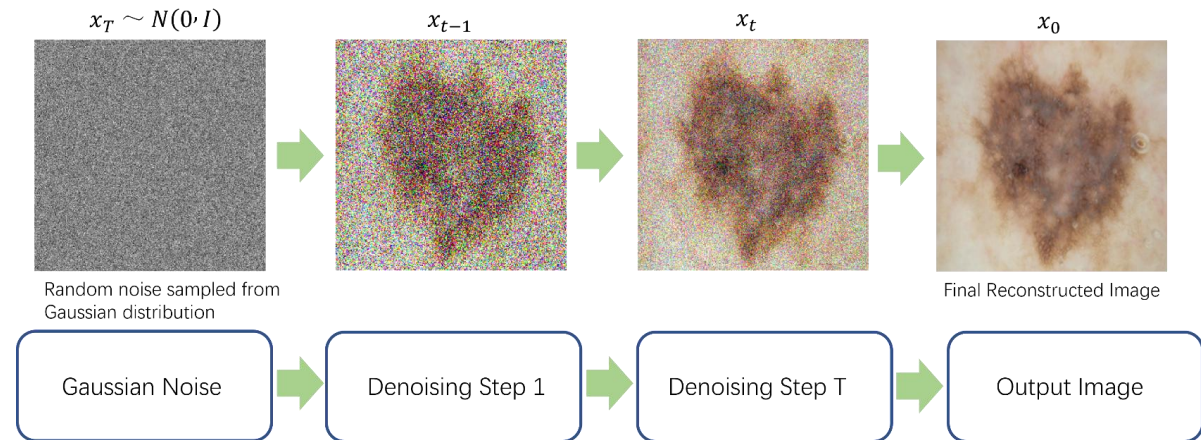
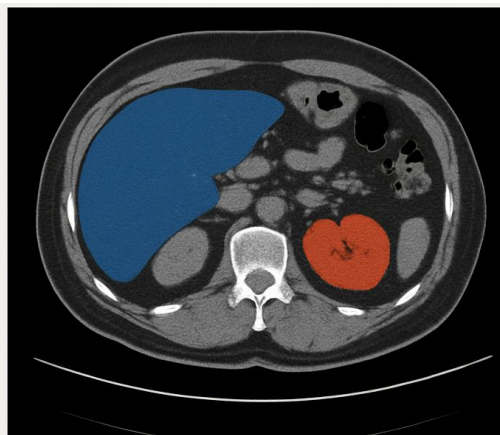
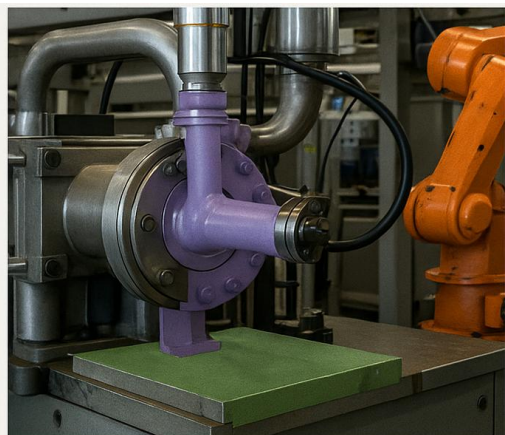
More recently, my work has involved introducing artificial intelligence into industrial production lines, particularly for quality inspection, as well as the development, integration, and maintenance of production-line equipment.

My work covers image segmentation, object detection, synthetic data generation, and improving model generalizability across different datasets and production environments.



我在學校的研究方向是醫療影像的分割。
近期的部分是將人工智能導入工業生產線上的品質檢測。
當然範圍包含，分割、檢測、數據生成、數據泛化性

<https://github.com/kancheng/happykoala>



Engineering Skills

Hardware

Hardware Experience

- Linux / Ubuntu server setup
- Computer & GPU installation
- Industrial camera & lighting setup
- Network devices, RJ45 / TCP/IP
- Equipment connection & verification

Troubleshooting Process

- Check power and network first
- Verify camera and lighting
- Confirm image quality
- Test software and hardware settings
- Find root cause step by step

Manufacturing Proof

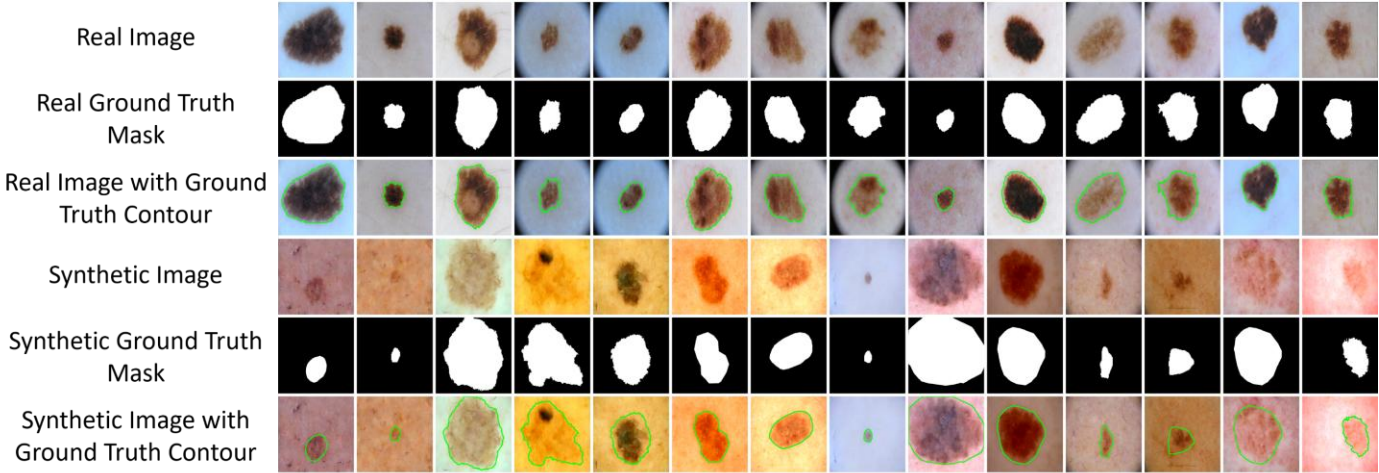
- Industrial image inspection systems
- Production-line workflow simulation
- Quality inspection process
- ONNX deployment
- System integration and testing

Software + Hardware + Manufacturing

Work and Research Experience

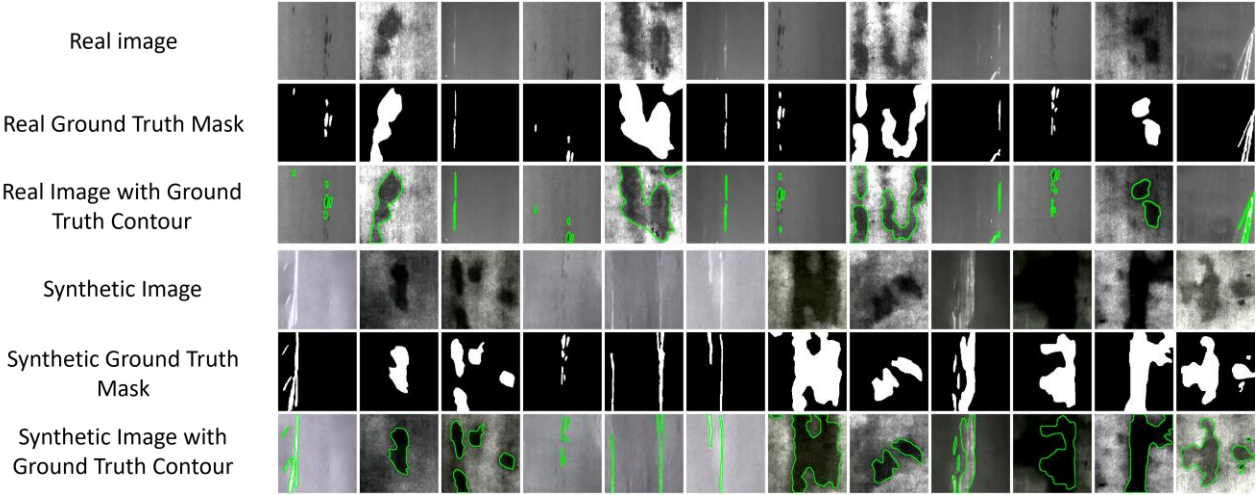
Medical

醫療



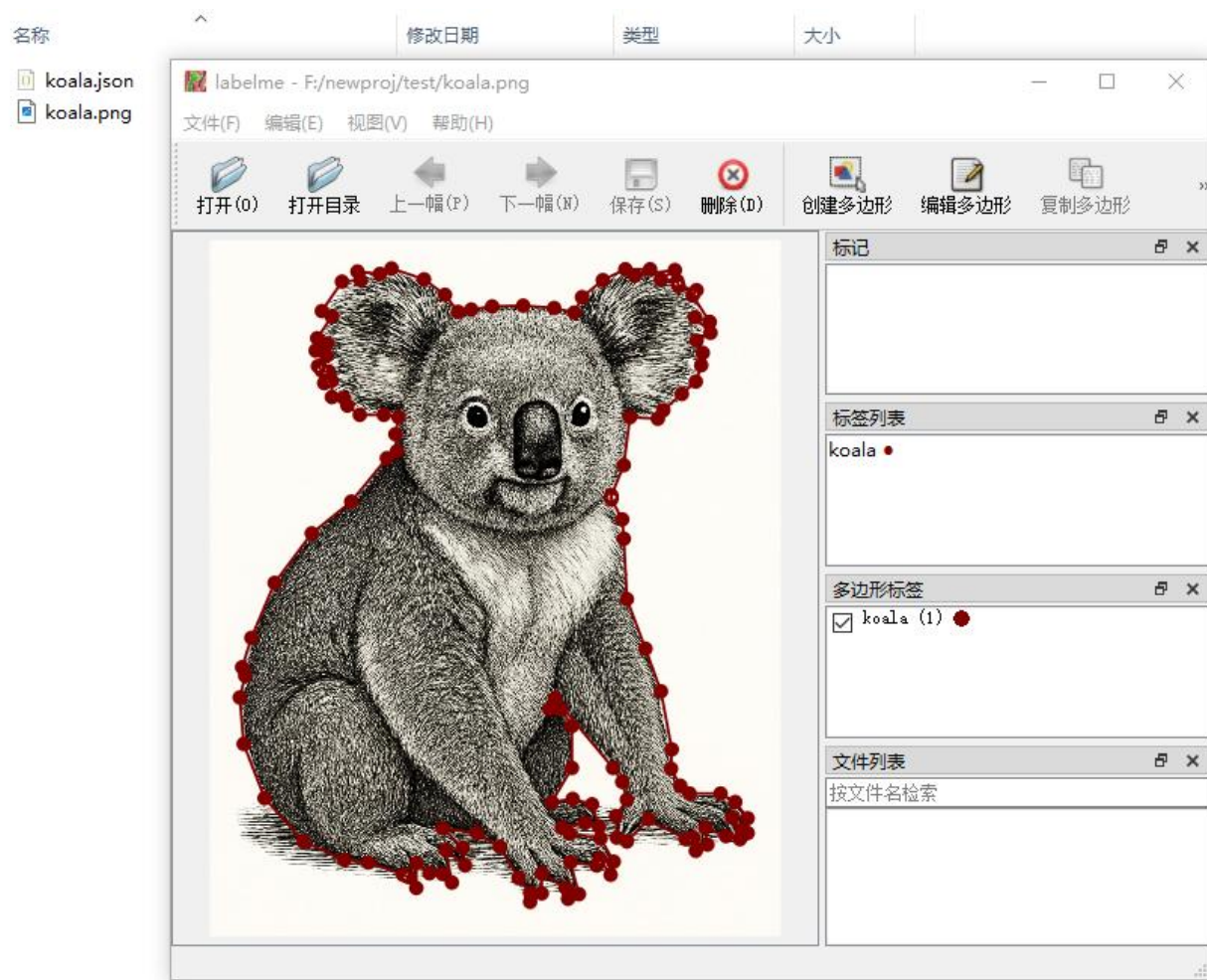
Industrial

工業



開源貢獻

<https://github.com/wkentaro/labelme>
<https://github.com/kancheng/labelme>



Labelme is a graphical image annotation tool inspired by <http://labelme.csail.mit.edu>. It is written in Python and uses Qt for its graphical interface.

Looking for a simple install without Python or Qt? Get the standalone app at labelme.io.



VOC dataset example of instance segmentation.



Other examples (semantic segmentation, bbox detection, and classification).



Various primitives (polygon, rectangle, circle, line, and point).

數據標註跟基本分割

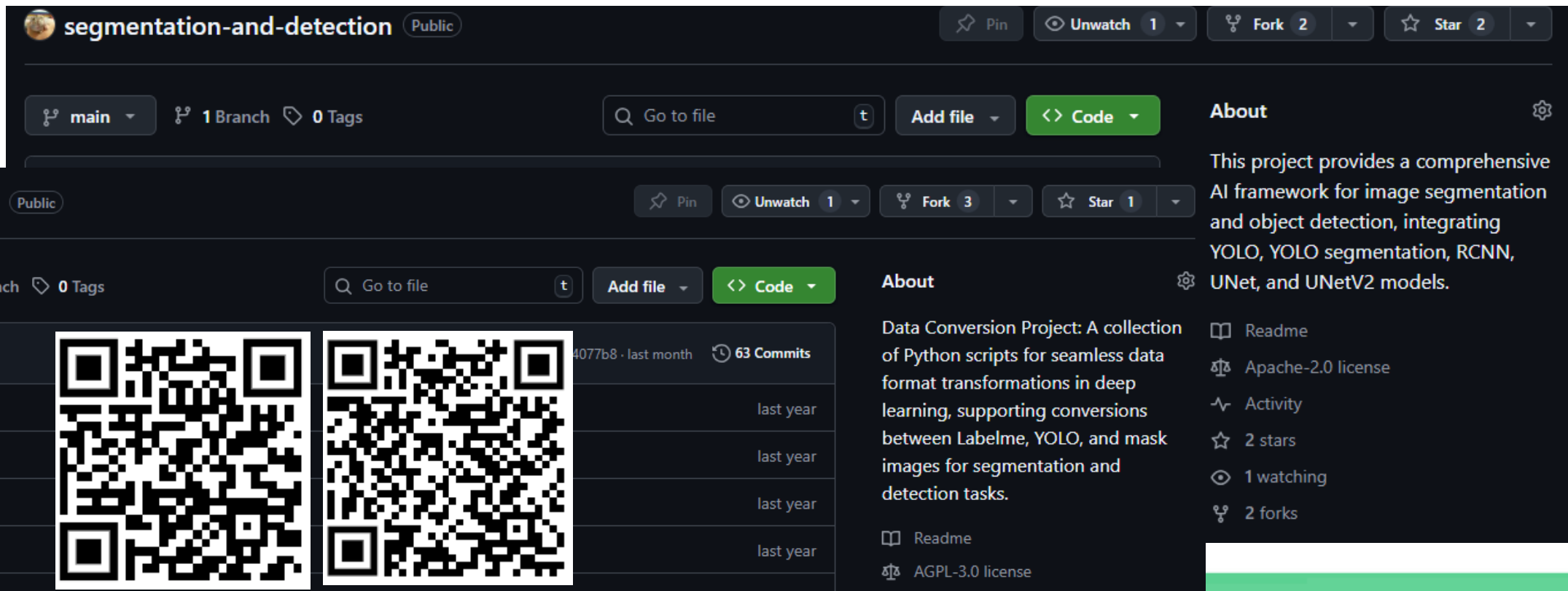


<https://github.com/kancheng/coco128-seg-labelme-format>

數據分割的部分，可自己訓練模型。EX: Labelme、YOLO、Masks 格式
在使用 Labelme 工具標註完後，轉換成 YOLO、Masks 格式。然後切記要在訓練中加入OK 圖。

<https://github.com/kancheng/data-conversion>

<https://github.com/kancheng/segmentation-and-detection>



segmentation-and-detection Public

main 1 Branch 0 Tags

Go to file Add file Code

About

This project provides a comprehensive AI framework for image segmentation and object detection, integrating YOLO, YOLO segmentation, RCNN, UNet, and UNetV2 models.

Readme Apache-2.0 license Activity 2 stars 1 watching 2 forks

data-conversion Public

main 1 Branch 0 Tags

Go to file Add file Code

About

Data Conversion Project: A collection of Python scripts for seamless data format transformations in deep learning, supporting conversions between Labelme, YOLO, and mask images for segmentation and detection tasks.

Readme AGPL-3.0 license

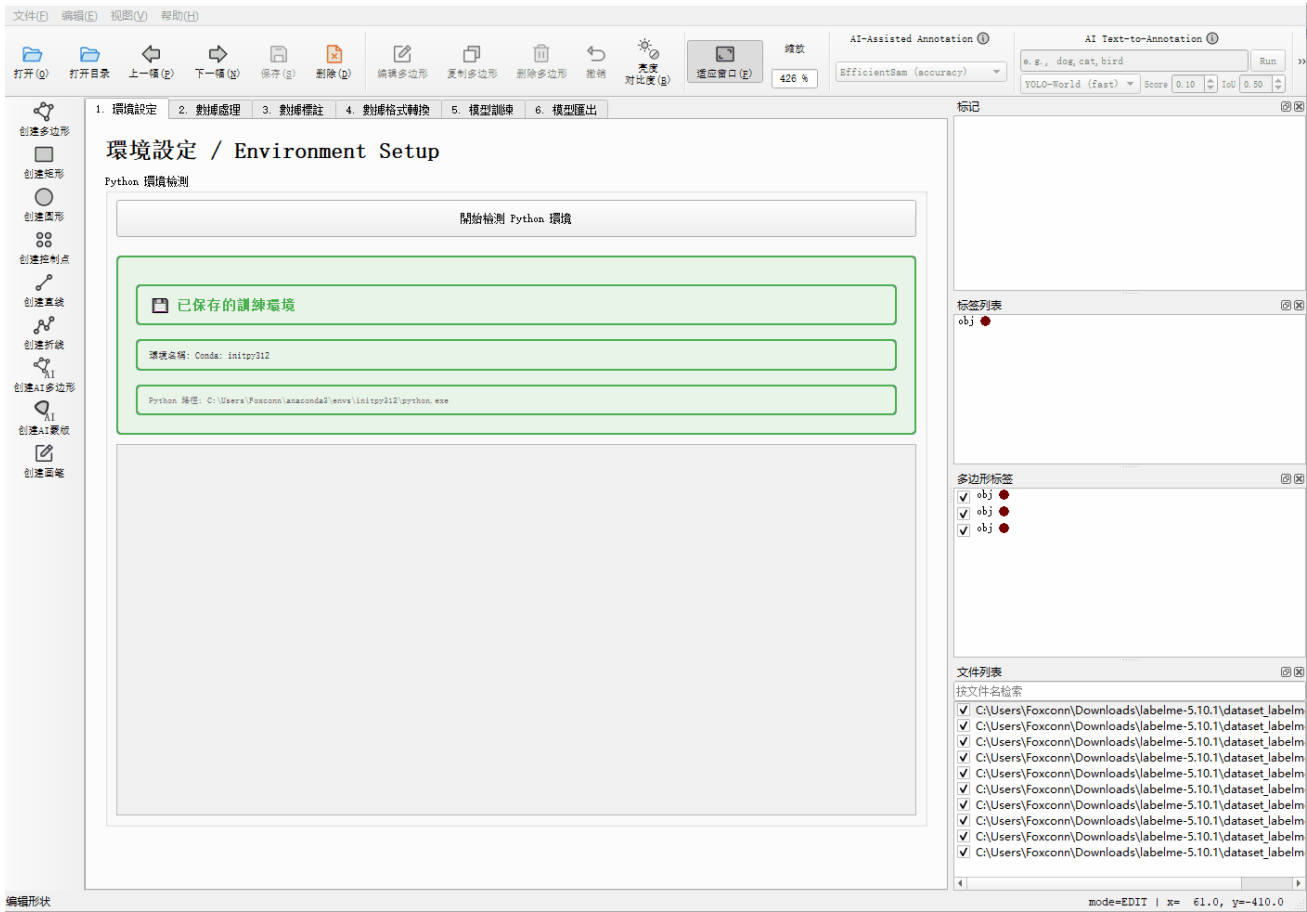
Commit	Author	Message	Time
4077b8	kancheng	Added	last month
			63 Commits
			last year
			last year
			last year
			last year

datasets docu function outputs

PyQT - Cappybara

<https://github.com/kancheng/labelme/tree/dev/capybara>

根據 Labelme 進行為基礎，徹底改寫為 GUI 操作的影像分割與檢測訓練工具。
其功能包含有數據轉換、增強、訓練、匯出為 onnx.



C# - YoloIndustrySegmentation

YOLO 模型訓練為 onnx，並撰寫一個基於 C# 的 GUI 檢測方案。



<https://github.com/kancheng/YoloIndustrySegmentation>

C#

Form1

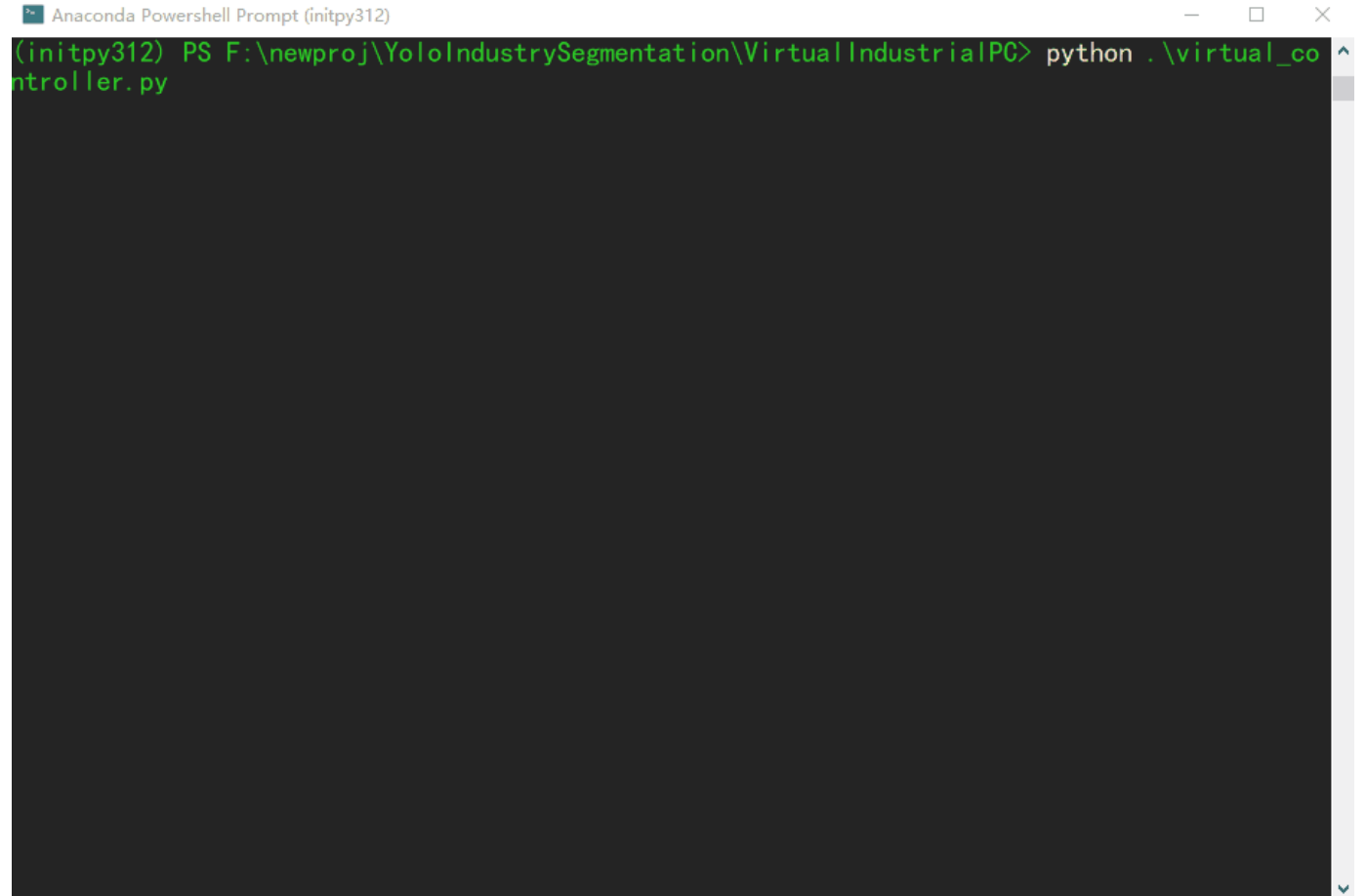
基本运动

			速度设置	距离设置					
X正	X反	X停止	速度	10000	距离	10000	目标位置		状态
Y正	Y反	Y停止	速度	10000	距离	10000	目标位置		状态
Z正	Z反	Z停止	速度	10000	距离	10000	目标位置		状态
X轴回原点	Y轴回原点	Z轴回原点	初始化系统		关闭系统		回原点		
			等待初始化.						
停止	急停	正向联动	负向联动		直线插补				

用户: 逻辑位置: 实际位置:

Python - YoloIndustrySegmentation

Production Line Simulation- Python 該項目有一個模擬生產線行為的 Python。



```
Anaconda Powershell Prompt (initpy312)
(initpy312) PS F:\newproj\YoloIndustrySegmentation\VirtualIndustrialPC> python .\virtual_co
ntroller.py
```



<https://github.com/kancheng/YoloIndustrySegmentation>

CV-YOLO & RF-DETR

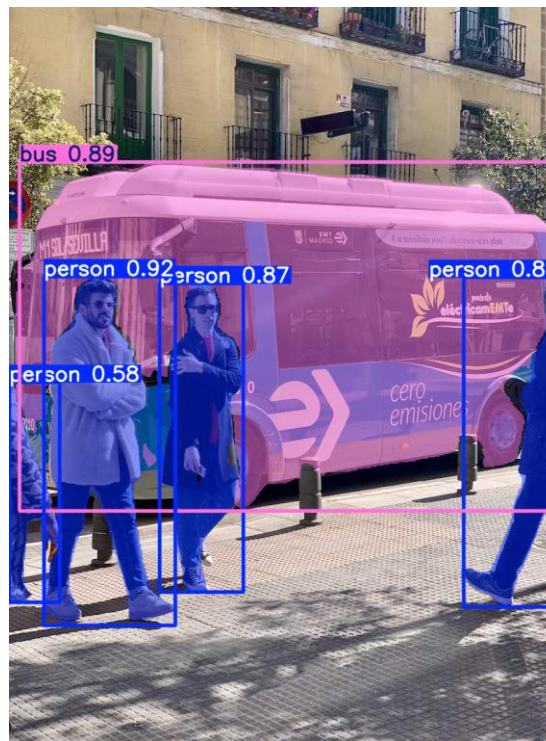
RAW



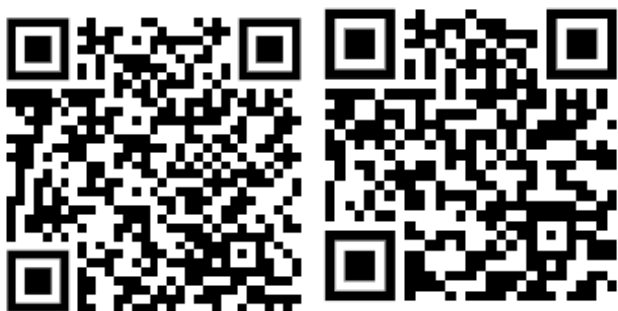
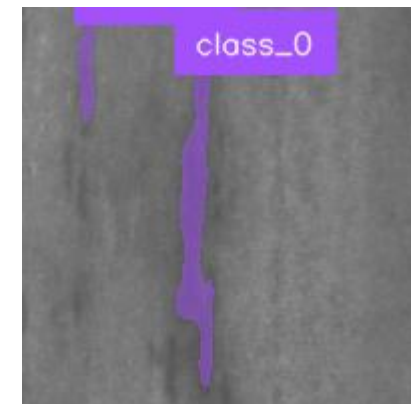
RF-DETR



YOLO



RF-DETR 金屬分割



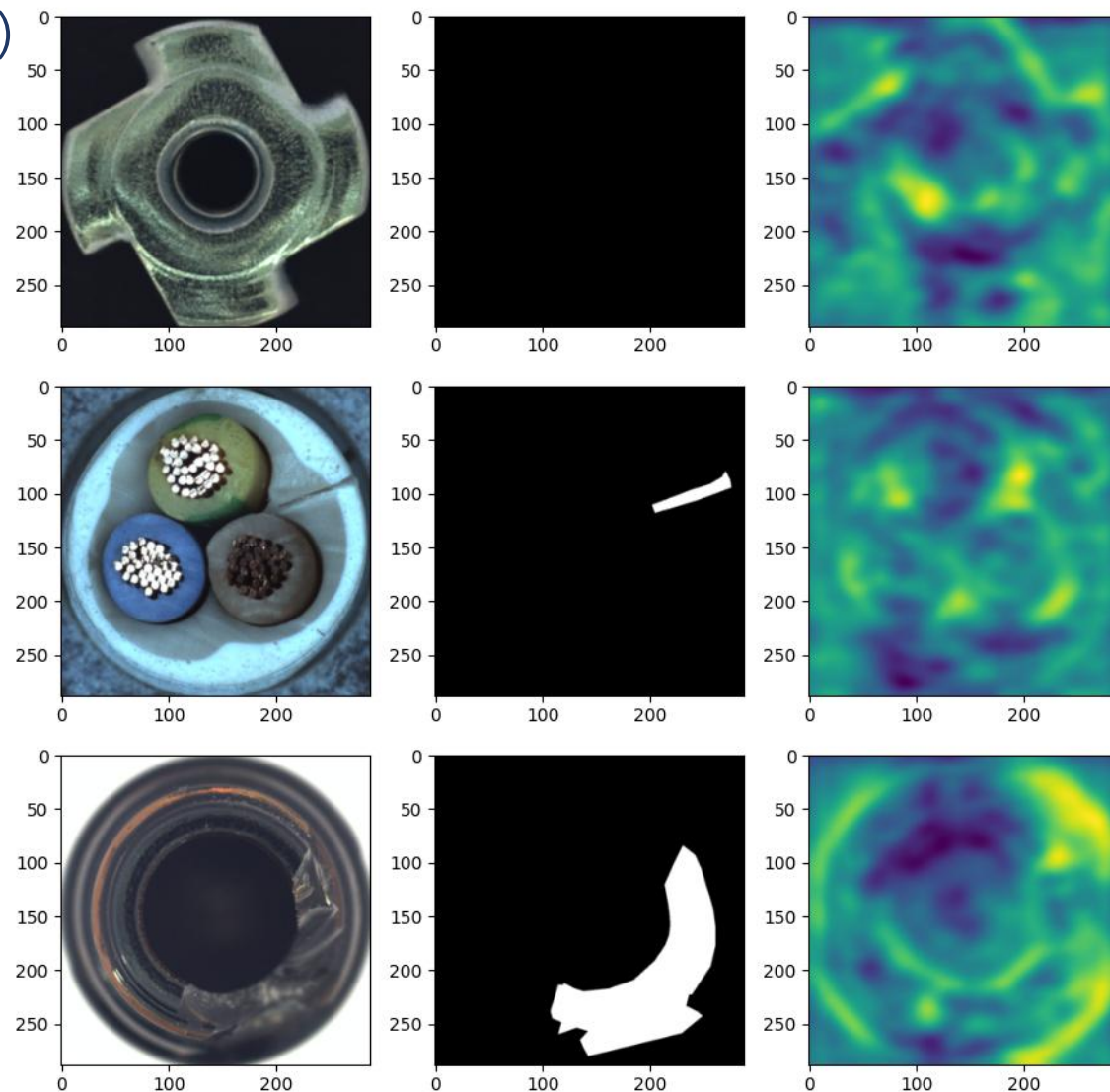
<https://github.com/kancheng/rfdetr-seg-train-predict>
<https://github.com/kancheng/yolo-vs-detr-demo>

Unsupervised Anomaly Detection

非監督式異常檢測 (Unsupervised Anomaly Detection)

- SimpleNet (CVPR 2023)
- SuperSimpleNet (ICPR 2024)

<https://arxiv.org/pdf/2408.03143>
<https://github.com/blaz-r/supersimplenet>
<https://arxiv.org/abs/2303.15140>
<https://github.com/DonaldRR/SimpleNet>



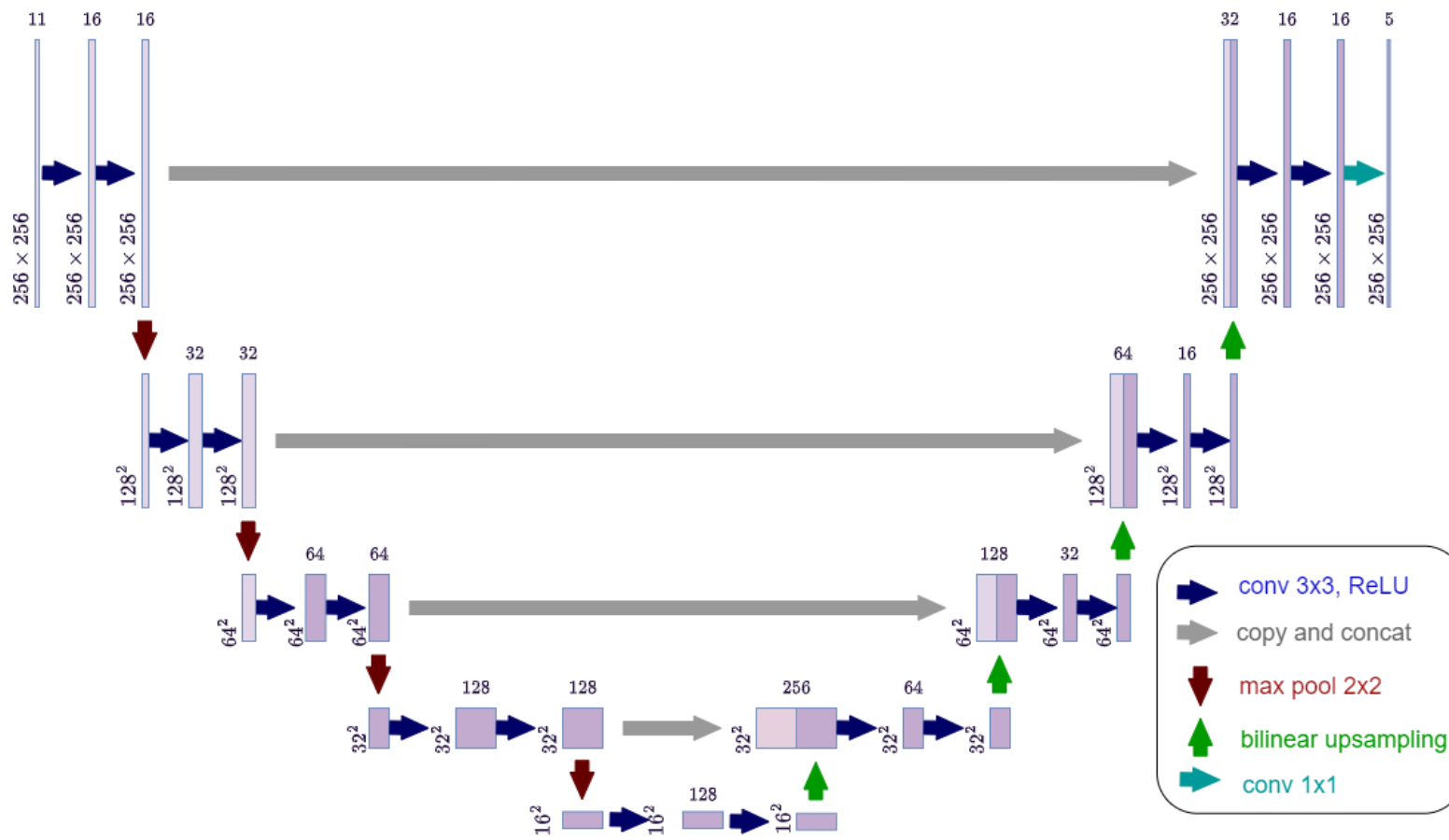
分割模型

UNet, U2Net, TranUnet, VMUnet, ResUnet, UNet++, Unet++

最早於 2015 年由 Olaf Ronneberger 等人提出，其論文名為《U-Net: Convolutional Networks for Biomedical Image Segmentation》。

結構呈現對稱的 U 型，主要由兩個部分組成：

1. 編碼器 (Encoder / Contracting Path)
2. 解碼器 (Decoder / Expansive Path)



初步消融實驗

UNet, U2Net,TranUnet, VMUnet, ResUnet,UNet+ +,Unet+ + +

Type	mIOU	DSC	Accuracy
UNET	0.6877	0.8150	0.9190
VMUNET	0.7944	0.8854	0.9443
U2NET	0.7296	0.8437	0.9232
UNET+ +	0.5968	0.7475	0.8682
UNET+ + +	0.7177	0.8356	0.9246
VMUNETV2	0.7812	0.8771	0.9403
HVMUNET	0.7990	0.8883	0.9443
TRANSUNET	0.6277	0.7713	0.8982
RESUNET	0.6061	0.7547	0.8861
RESUNET+ +	0.6677	0.8007	0.9090

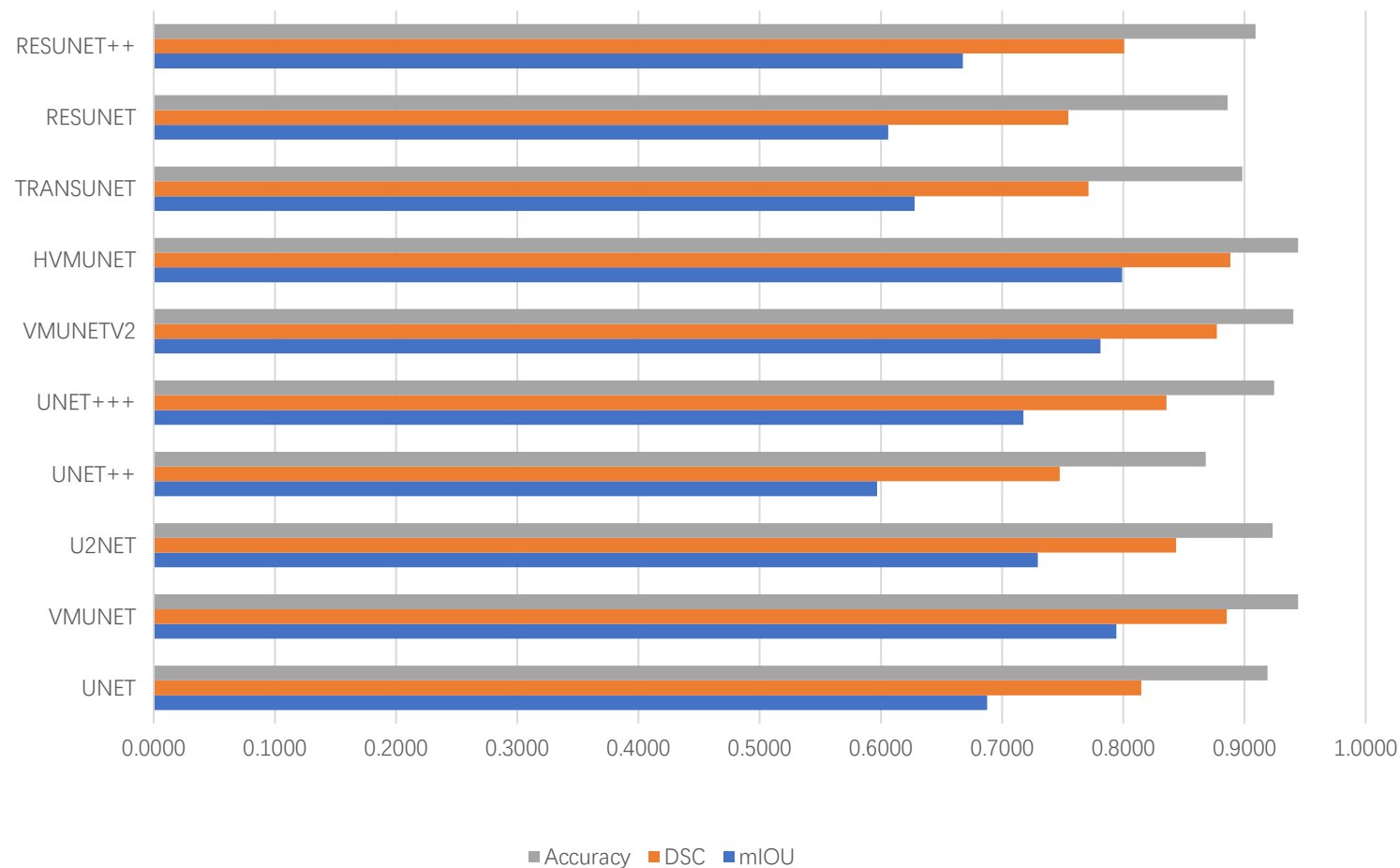
基礎模型，表現穩定
但非最佳。

優秀表現，三項指標
皆在前列。

所有指標皆為最佳或
並列最佳，表現最強。

初步消融實驗

UNet, U2Net, TranUnet, VMUnet,
ResUnet, UNet++, UNet+++, UNet++



mIOU (平均交併比)：評估預測區域與真實區域的重疊程度。數值越高表示模型預測越準確。

DSC (Dice 系數)：衡量兩區域相似性的指標，常用於醫療影像分割。越接近 1 越好。

Accuracy (準確率)：預測正確像素所佔總像素的比例。對於背景主導的資料集，準確率容易偏高，但不一定能真實反映分割品質。

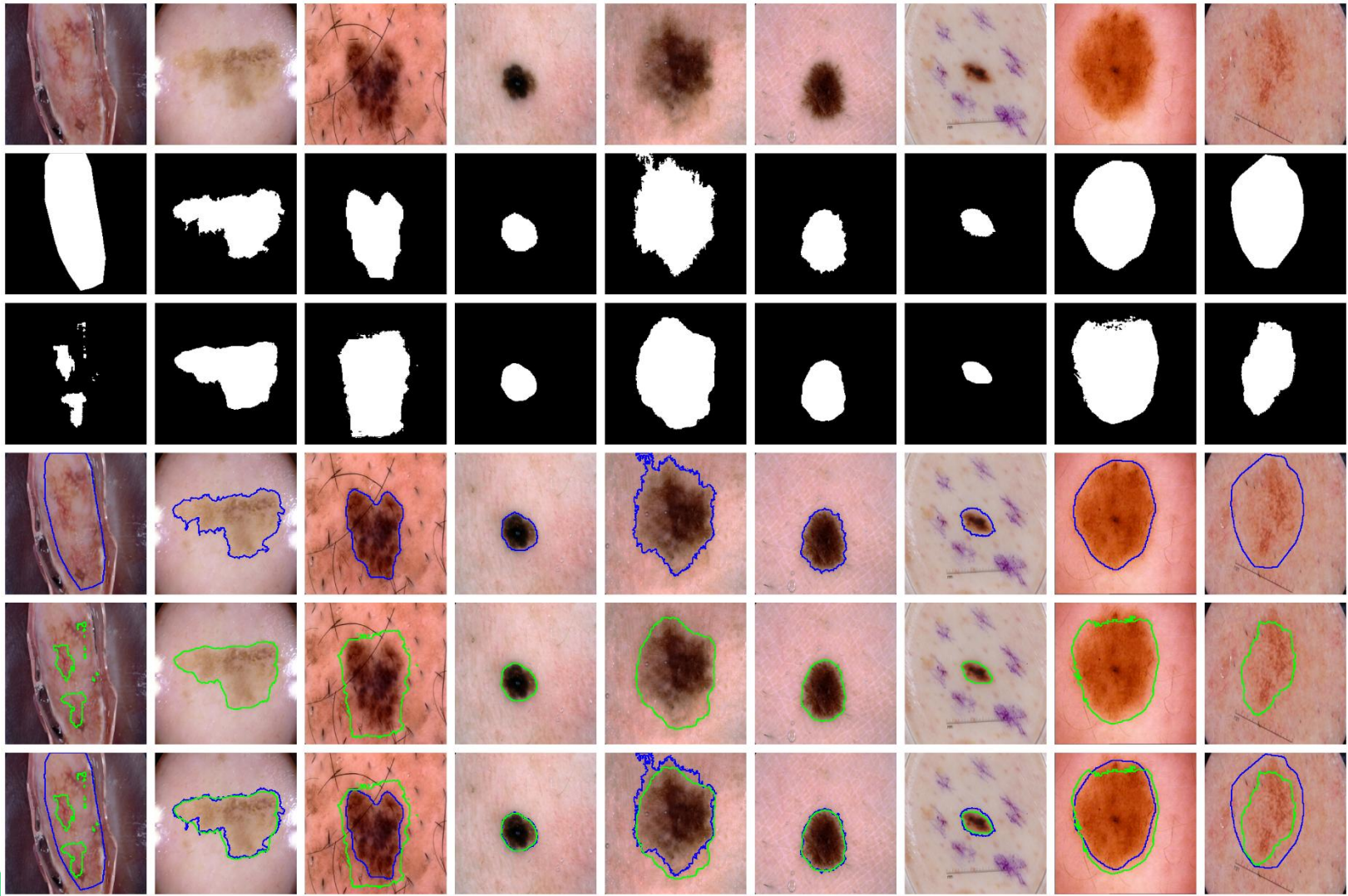
實驗呈現

UNET

原始影像
Image

標註真值
ground truth

模型預測結果
prediction



實驗呈現

VMUNET

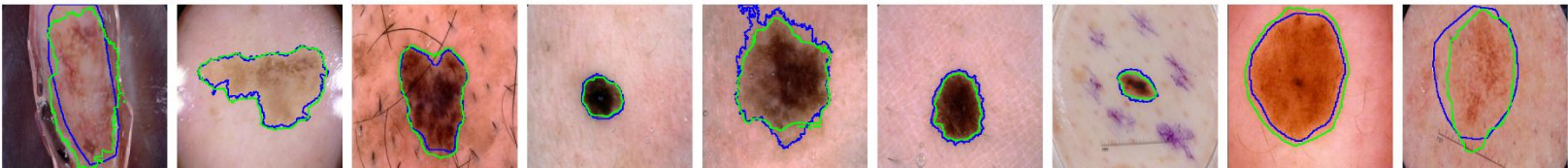
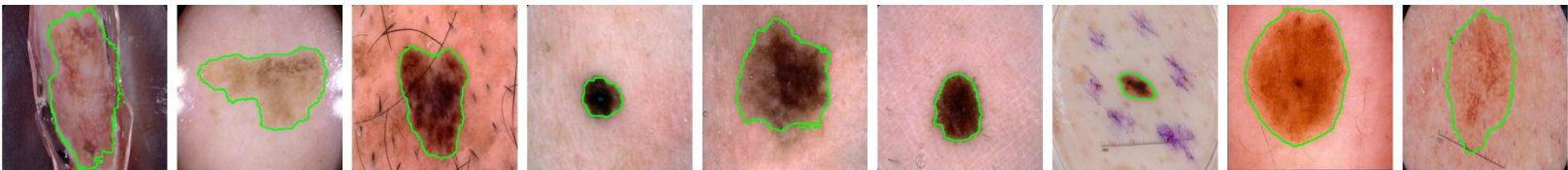
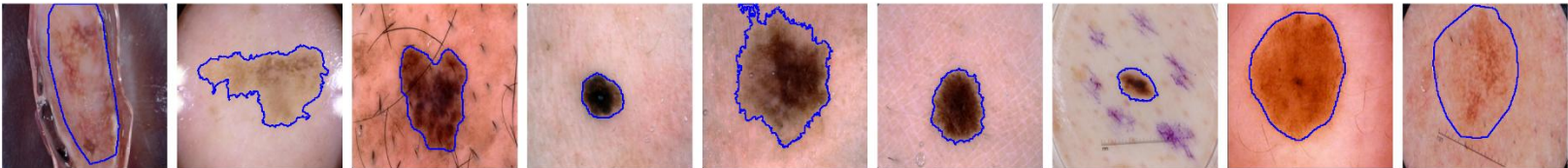
原始影像
Image



標註真值
ground truth

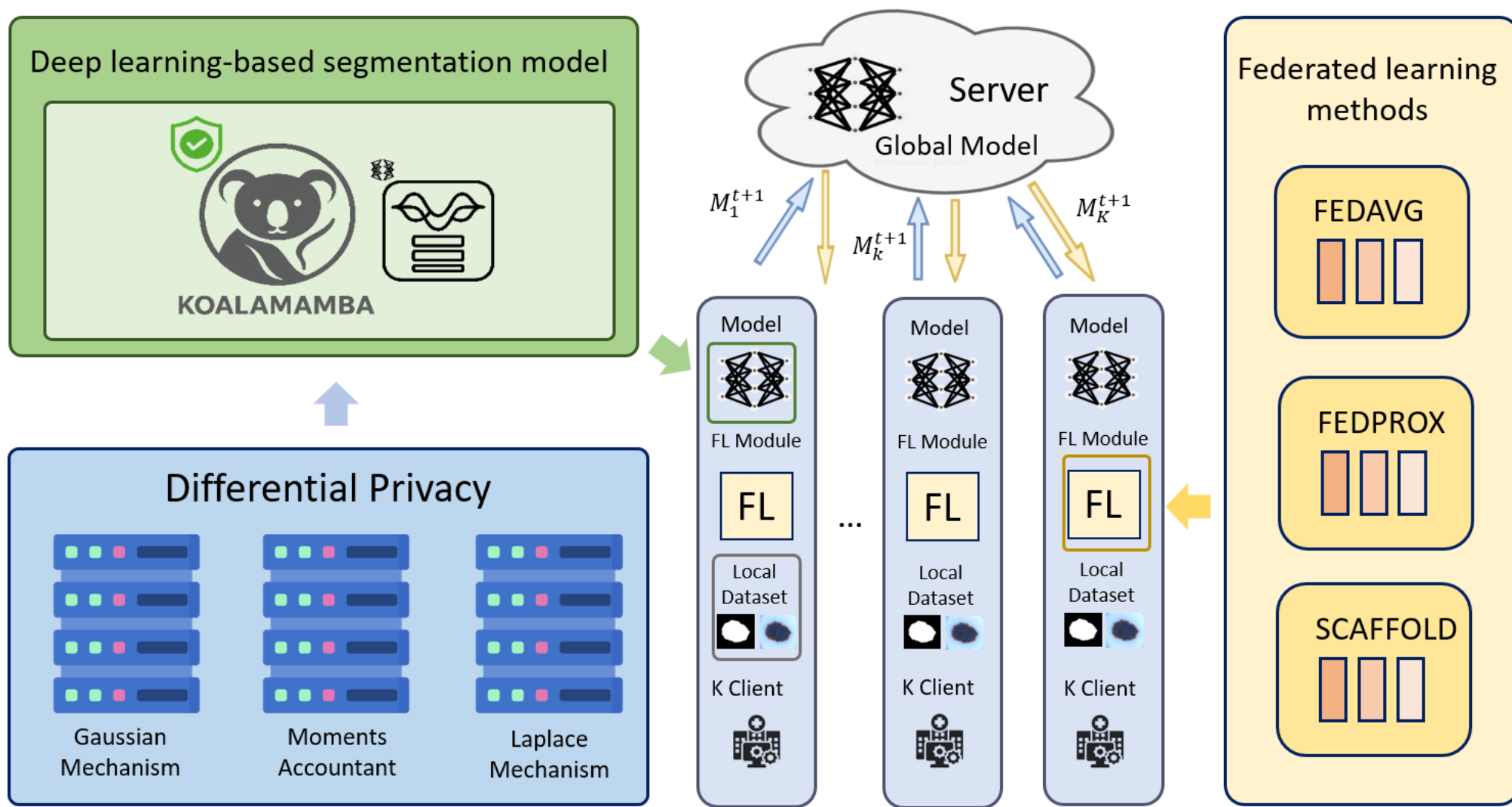


模型預測結果
prediction



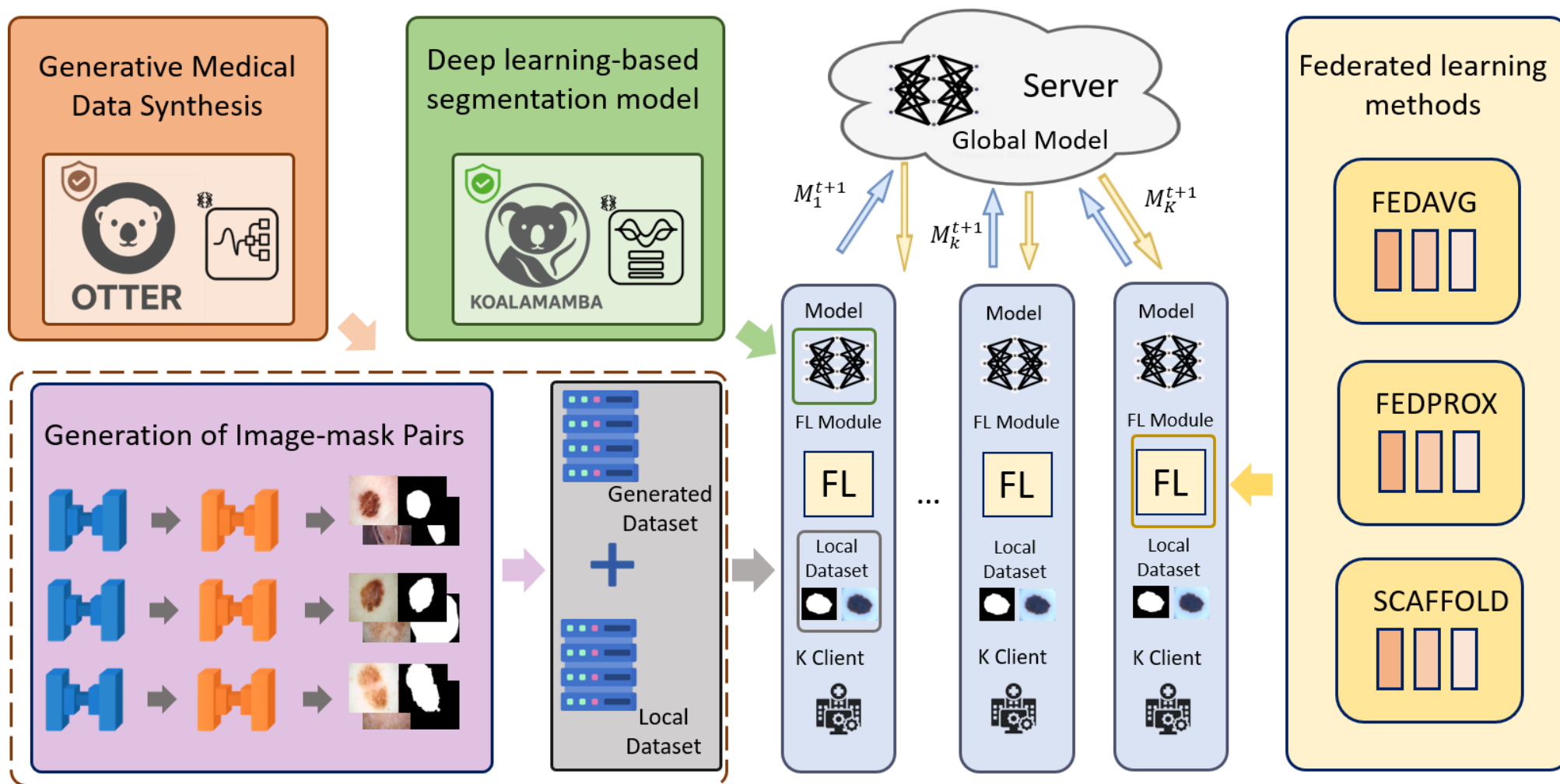
發表

1. **Haocheng Kan**, Yuesheng Zhu, Guibo Luo and Hanwen Zhang. OTTER: Optimized Training with Trustworthy Enhanced Replication via Diffusion and Federated VMUNet for Privacy-Aware Medical Segmentation. [C]//The 2025 International Conference on Information and Communications Security (ICICS). 2025. (CCF-C)
2. **Haocheng Kan**, Yuesheng Zhu, Guibo Luo and Mingpei Cao. KoalaMamba: A Federated Method with Vision Mamba UNet for Privacy-Preserving Medical Segmentation. [C]//WCCI 2026. (CCF-C)



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結論發表

醫療

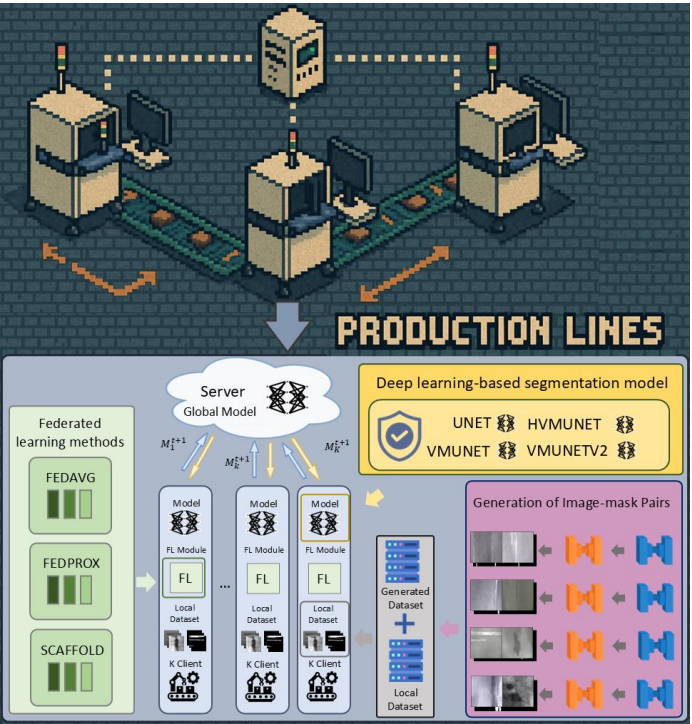
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工業

投稿中 ...

RABBIT: A Privacy-Aware Diffusion and VMUNet-Based Segmentation Framework for Multi-Factory Industrial Defects



End

Linux 底層開發、嵌入式設備、ONNX、DINOv3

<https://github.com/facebookresearch/dinov3>

